

L42 — Hash Tables and Hash Functions

Unit: 3 | Chapter: Hashing | BT Level: BT3 | CO: CO2

Learning Objectives

- Explain the concept of hashing and hash tables
 - Design and evaluate good hash functions
 - Analyze expected time complexity of hash table operations
 - Implement a basic hash table in Python
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1. Motivation

Problem: Store and retrieve data by key in $O(1)$ time.

- Array: $O(1)$ access by index, but index must be an integer $0..n-1$
- Linked list: $O(n)$ search
- BST: $O(\log n)$ search

Hash table: Maps arbitrary keys (strings, numbers, objects) to array indices using a **hash function** → average $O(1)$ for all operations.

2. Hash Table Concept

Key Space		Hash Function		Array (Table)
"Alice"	→	$h(\text{"Alice"}) = 2$	→	index 2
"Bob"	→	$h(\text{"Bob"}) = 5$	→	index 5
"Charlie"	→	$h(\text{"Charlie"}) = 2$	→	index 2 (COLLISION!)

Table: `[_ , _ , "Alice"/"Charlie", _ , _ , "Bob", ...]`

Collision: Two different keys map to the same index. Must be handled (see L43).

3. Hash Function Properties

A good hash function should:

1. Be **deterministic**: same key always gives same hash
 2. Be **fast to compute**: $O(1)$ or $O(\text{key length})$
 3. Be **uniform**: distribute keys evenly across all slots
 4. **Minimize collisions**: different keys should hash to different slots
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4. Common Hash Functions

4.1 Division Method

$$h(k) = k \bmod m$$

- m = table size; best when m is a prime not close to a power of 2

```
def hash_division(key, m):  
    return key % m
```

4.2 Multiplication Method

$$h(k) = \lfloor m \cdot (k \cdot A \bmod 1) \rfloor \text{ where } A \approx (\sqrt{5} - 1)/2 \approx 0.618 \text{ (Knuth's constant)}$$

```
import math  
  
def hash_multiplication(key, m, A=0.618033988749895):  
    return int(m * ((key * A) % 1))
```

4.3 String Hashing (Polynomial Rolling)

```
def hash_string(s, m, base=31):  
    """Polynomial rolling hash. O(|s|)"""  
    h = 0  
    for char in s:  
        h = (h * base + ord(char)) % m  
    return h
```

Better with a large prime modulus to minimize collisions:

```
def hash_string_robust(s):  
    """Robust string hash using large prime modulus."""  
    MOD = (1 << 61) - 1    # Mersenne prime  
    BASE = 131  
    h = 0  
    for char in s:  
        h = (h * BASE + ord(char)) % MOD  
    return h
```

4.4 Python's Built-in Hash

```
hash("hello")          # Returns integer (varies per Python session)  
hash(42)                # 42  
hash((1, 2, 3))         # Tuple hash  
hash([1, 2])            # TypeError: unhashable type: list
```

Python uses SipHash (with random seed per process for security).

5. Hash Table Implementation

```
class HashTable:  
    def __init__(self, capacity=16):
```

```

        self.capacity = capacity
        self.size = 0
        self.buckets = [[] for _ in range(capacity)]    # Chaining

def _hash(self, key):
    return hash(key) % self.capacity

def put(self, key, value):
    """Insert or update key-value pair. Average O(1)"""
    idx = self._hash(key)
    bucket = self.buckets[idx]

    for i, (k, v) in enumerate(bucket):
        if k == key:
            bucket[i] = (key, value)    # Update existing
            return

    bucket.append((key, value))        # New entry
    self.size += 1
    if self.size / self.capacity > 0.75:
        self._resize()

def get(self, key):
    """Retrieve value for key. Average O(1)"""
    idx = self._hash(key)
    for k, v in self.buckets[idx]:
        if k == key:
            return v
    raise KeyError(key)

def delete(self, key):
    """Delete key. Average O(1)"""
    idx = self._hash(key)
    bucket = self.buckets[idx]
    for i, (k, v) in enumerate(bucket):
        if k == key:
            bucket.pop(i)
            self.size -= 1
            return
    raise KeyError(key)

def _resize(self):
    """Double capacity when load factor exceeds 0.75. O(n)"""
    old_buckets = self.buckets
    self.capacity *= 2
    self.size = 0
    self.buckets = [[] for _ in range(self.capacity)]

    for bucket in old_buckets:
        for key, value in bucket:
            self.put(key, value)

def __contains__(self, key):

```

```
idx = self._hash(key)
return any(k == key for k, _ in self.buckets[idx])
```

6. Load Factor

$$\alpha = \frac{n}{m}$$

where n = number of entries, m = number of slots.

- $\alpha < 0.75$: Good performance (Python dict target)
 - $\alpha > 1$: Many collisions, degraded performance
 - When α exceeds threshold $\rightarrow$ **rehash** (double capacity, reinsert all)
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7. Expected Time Complexity

Operation	Average Case	Worst Case
Search	$O(1 + \alpha) \approx O(1)$	$O(n)$ — all keys collide
Insert	$O(1)$ amortized	$O(n)$
Delete	$O(1 + \alpha) \approx O(1)$	$O(n)$
Space	$O(n + m)$	$O(n + m)$

With good hash function and α kept $<$ constant $\rightarrow$ **$O(1)$ expected.**

8. Applications

Application	Hash Table Use
Python dict, set	Key-value store, membership test
Database indexing	Hash index for equality queries
Compiler symbol table	Variable name $\rightarrow$ memory address
Cache (LRU Cache)	Key $\rightarrow$ cached value
Deduplication	Detect duplicate records
Frequency counting	Count occurrences of elements
Cryptographic hashing	MD5, SHA-256 (for data integrity, not tables)

9. Python dict and set

Python's built-in dict and set are backed by a highly optimized hash table:

```
# Dict operations are O(1) average
d = {}
d['key'] = 'value'      # O(1) insert
x = d['key']            # O(1) lookup
del d['key']            # O(1) delete
'key' in d              # O(1) membership
```

```
# Set for deduplication
seen = set()
unique = [x for x in items if not (x in seen or seen.add(x))]

# Frequency count
from collections import Counter
freq = Counter("abracadabra")
print(freq)    # {'a': 5, 'b': 2, 'r': 2, 'c': 1, 'd': 1}
```

Summary

- Hash table: maps keys to array indices via a **hash function** → $O(1)$ average operations.
 - Good hash function: uniform distribution, fast, deterministic.
 - **Load factor $\alpha = n/m$** : keep below 0.75; rehash when exceeded.
 - Collisions are inevitable — handled by chaining (L43) or open addressing (L43).
 - Python's `dict` and `set` are practical hash table implementations.
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References

- Cormen et al., *Introduction to Algorithms*, Ch. 11 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 3.7 (Springer, 2020)